Output tables for 1xN statistical comparisons.

March 10, 2025

1 Average rankings of Friedman test

Average ranks obtained by each method in the Friedman test.

Algorithm	Ranking
BHCVRP-Java-anterior	2
BHCVRP-Java-actual	2
BHCVRP-Python-actual	2

Table 1: Average Rankings of the algorithms (Friedman)

Friedman statistic (distributed according to chi-square with 2 degrees of freedom): -0.
P-value computed by Friedman Test: 1.

2 Post hoc comparison (Friedman)

P-values obtained in by applying post hoc methods over the results of Friedman procedure.

i	algorithm	$z = (R_0 - R_i)/SE$	p	Holm	Finner	Li
2	BHCVRP-Java-anterior	0	1	0.025	0.025321	0
1	BHCVRP-Python-actual	0	1	0.05	0.05	0.05

Table 2: Post Hoc comparison Table for $\alpha = 0.05$ (FRIEDMAN)

Holm's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.025.
Finner's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.025321.
Li's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.

3 Adjusted P-Values (Friedman)

Adjusted P-values obtained through the application of the post hoc methods (Friedman).

i	algorithm	unadjusted p	p_{Holm}
1	BHCVRP-Java-anterior	1	2
2	BHCVRP-Python-actual	1	2

Table 3: Adjusted p-values (FRIEDMAN) (I)

i	algorithm	unadjusted p	p_{Finner}	p_{Li}
1	BHCVRP-Java-anterior	1	1	1
2	BHCVRP-Python-actual	1	1	1

Table 4: Adjusted p-values (FRIEDMAN) (II)